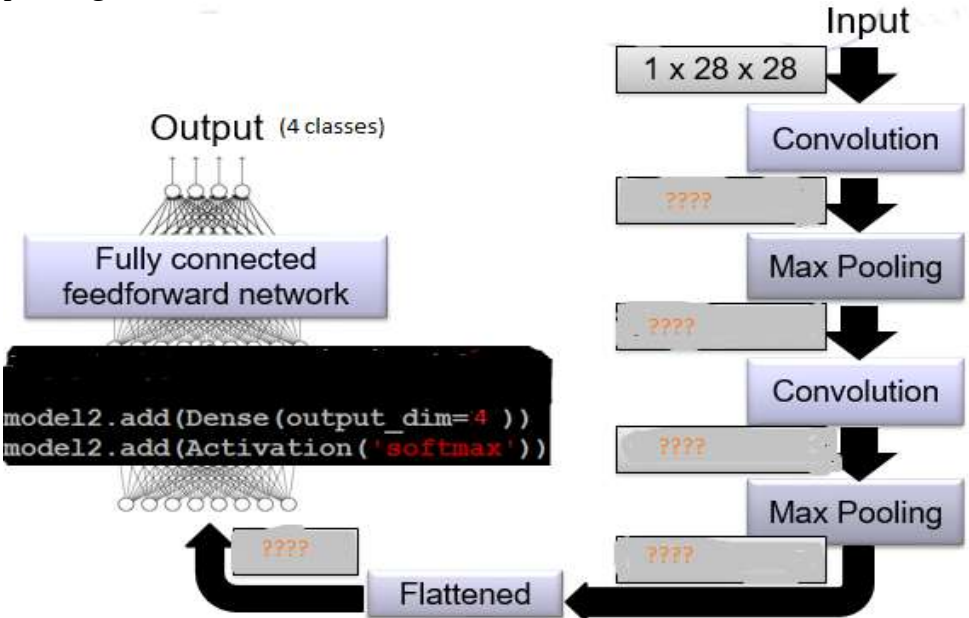




BRAC UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science / CSE Program
Final Exam, Spring 2022

Course: CSE437 (Data Science: Coding With Real World Data), Section-1
Instructor: Dr. Md. Golam Rabiul Alam
Full Marks: 25
Time: 2 hours

Note: There are Five questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1.	Analyze the output size of each of the CNN layers of the following CNN model. Furthermore, mention the number of learning parameters required to learn in 1st convolution, 2nd convolution, and fully connected layers. Consider the following values of the CNN parameters: Number of filters (for each of the convolution operations) = 8 Filter size = 2 x 2 (for each of the convolution operations) stride=1 padding='valid' Pool size = 2 x 2 (for each of the pooling operations) stride=2 padding='valid' 	[CO2, C4, Marks:5]
2.	Apply K-means clustering algorithm to partition the following points into 3 clusters: A(2, 10), B(2, 5), C(8, 4), D(5, 8), E(7, 5), F(6, 4), G(1, 2), H(4, 9) Consider the points A(2, 10), D(5, 8) and G(1, 2) as the initial cluster center. Use the Cosine Distance function to measure the dissimilarities. Finally, show the clusters in a scatter plot of the points.	[CO2, C3, Marks:10]

3. The following figure represents a neural network with one hidden layer.

[CO3, C3,
Marks:10]

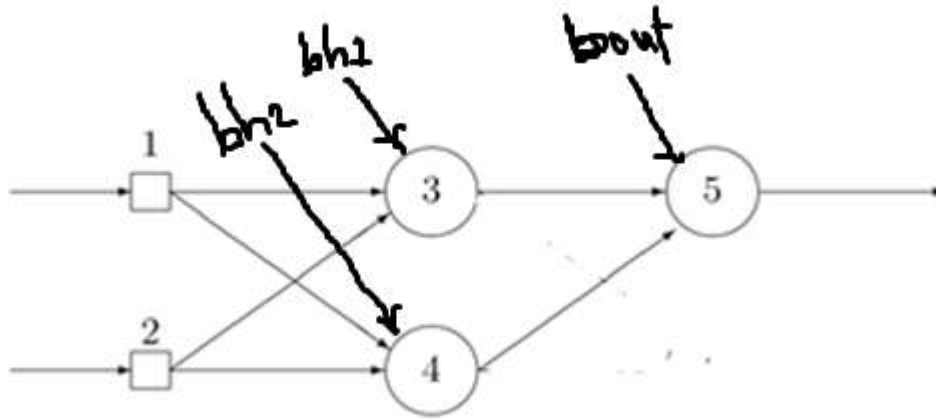


Figure 1: A neural network with two input layer neuron (1, 2), two hidden layer neuron (3, 4) and one output layer neuron (5).

A weight on connection between nodes i and j is denoted by w_{ij} , such as w_{13} is the weight on the connection between nodes 1 and 3. The following table lists all the initial weights in the network.

$w_{13} = -2$	$w_{35} = .1$
$w_{23} = .3$	$w_{45} = -.1$
$w_{14} = .4$	bh1=.5
$w_{24} = -.1$	bh2=.2
	bout=0.4

Each of the nodes 3, 4, and 5 uses the following sigmoid activation function: Each of the input nodes (1 and 2) can only receive binary values (either 0 or 1).

Consider the following dataset and **apply** the forward propagation and backward propagation (only one cycle or iteration) to show the **predicted output** and **updated weights** using batch processing of batch size 4. Consider the learning rate as 0.1

Input		Output
X1	X2	Y
1	1	1
1	0	0
0	1	0
0	0	1